

numbers. It was not until the 1980s, a decade after the US, that they introduced their first truly modern fighters, the MiG-29 and Sukhoi Su-27. These and other subsequent designs showed how brilliant Soviet scientists and engineers could be, but deep-seated problems remained. The Soviet Union was seriously behind the West in computing and microelectronics. And the need to operate such complex aircraft put a strain on every element of the Soviet system, from pilot training to ground maintenance.

The modern fighter pilot

The demands that the new generation of fighter aircraft made on pilots were not entirely novel. In a sense, the new airplanes were firmly in a tradition that went back to the World War II Spitfire and beyond, with the aircraft experienced as an extension of the pilot's body – pilots speak of “putting the fighter on” when they climb into the cockpit. The qualities required of a fighter pilot remained much the same as in an earlier era: natural aggression, physical fitness, good eyesight, spatial sense, all-around situational awareness, keen intelligence, and a cool head.

In some ways, modern fighter pilots enjoyed advantages their predecessors would have envied. Virtual-reality simulators largely ended the waste of young lives in training accidents. Overall, being a fighter pilot ceased to be the high-risk occupation that it was in the 1950s and 1960s. Accidental fighter loss rates halved in each decade from 1970 onward, with more reliable engines, better understanding of jet-aircraft design, and the introduction of computerized controls that to a large degree obviated pilot error. Devices such as terrain-following radar linked to an autopilot allowed aircraft to fly supersonic at altitudes as low as 200ft (60m), even over rough terrain, with reasonable safety.

There was no question, though, that the new aircraft imposed some unprecedented physical demands on the pilot. In earlier times, the main risk of high-speed maneuvers had been that they would put excessive strain on the airframe. In modern fighters, the risk was excessive strain on the pilot's physiology. The largest g-force that a person can withstand before blacking out is about 5g. But the pilot might be flying an aircraft capable of, say, 12-g maneuvers. Part of the elaborate gear in which a pilot now steps out to his aircraft is a g-suit; fitting tightly around the lower torso and upper legs, it constricts the bottom half of the body to keep blood in the upper body and brain. This can improve resistance up to 6 or 7g. Beyond this point, pilots have to resort to tensing and

McDonnell Douglas F-15 Eagle



ACTION STATIONS

On July 27, 1988, two F-15s fulfilled their originally intended fighter-interceptor role by intercepting two Soviet Tu-95 “Bear” long-range bombers off the east coast of Canada.

WHEN THE SOVIETS UNVEILED their MiG-25 interceptor in 1967, American defense experts put out an urgent call for a new air-superiority fighter of outstanding performance and agility to match it. The result was the F-15 Eagle fighter-interceptor, whose lightweight construction materials and powerful, specially developed Pratt & Whitney turbofan engines gave it an excellent thrust-to-weight ratio. In a vertical climb the Eagle could reach 29,000ft (8,850m) – the height of Everest – in under a minute.

Most F-15s, which entered service in 1974, are single-seaters, meaning that the pilot has to cope with target identification, spotting and tracking targets, deploying countermeasures against missile attack, and all the other complex business of modern warfare, as well as flying. The designers did an excellent job of making the pilot's task more manageable, and many of their innovations, such as HOTAS (“hands on throttle and stick”) became standard on all modern fighters. Later models (the F-15C and D) carried more advanced systems, including the improved APG-70 radar.

Although not as agile in a dogfight as some smaller, lighter aircraft, the F-15's unique ability to accelerate in vertical flight provided the perfect escape route if the pilot was ever in trouble. None has ever been lost in combat.

Rear-facing radar warning receiver

Electronic countermeasures (ECM) jammers

Rudder

Twin tailfin

Specifications (F-15A)

ENGINE 2 x 23,770lbst (10,780kcp) P&W F100 turbofan

WINGSPAN 42ft 8in (13m)

LENGTH 63ft 8in (19.4m)

WEIGHT 32,000lb (14,515kg) **CREW** 1

TOP SPEED 1,164mph (1,875kph) (Mach 2.5 plus)

ARMAMENT 1 x M-61A1 20mm canon; 4 x AIM-9L/M Sidewinder and 4 x AIM-7F/M Sparrow air-to-air missiles, or 8 x AIM-120 AMRAAMs

Missile rail

Starboard main landing gear